ROBUST DEEPFAKE DETECTION FOR NTIRE 2026

Bridging the Gap Between Lab Accuracy and "In-the-Wild" Robustness.

THE PROBLEM

- **The Current State:** Standard detectors achieve 99% accuracy on specific datasets (e.g., FaceForensics++) but fail catastrophically (drop to ~60%) on "unseen" generators.
- **The NTIRE 2026 Challenge:**
  - **Unknown Generators:** Must detect newer Diffusion models (Midjourney v6, Sora, Flux) not just older GANs.
  - **Degradations:** Input images are heavily compressed (social media quality), blurred, or resized.
- **The Cybersecurity Threat**: Attackers use these "invisible" fakes for bypass attacks on e-KYC (Identity Verification) systems.

CORE ANALYSIS: MULTI-MODAL ANALYSIS

We cannot trust pixels alone. We need to look at the image in two domains simultaneously.

- **Spatial Domain (RGB):** What the human eye sees (visual artifacts, blending errors).
- **Frequency Domain (DCT/FFT):** What the machine sees (checkerboard patterns, upsampling fingerprints).
- **Hypothesis:** Even if an AI generates a visually perfect face, the mathematical process of generating pixels leaves a distinct fingerprint in the frequency spectrum that survives compression.

THE HYBRID-SWIN NETWORK

Architecture Overview:

- **Stream A (Spatial):** Swin Transformer V2
  - State-of-the-art for capturing long-range dependencies (global consistency) in high-resolution images.
- **Stream B (Frequency):** F3-Net (Frequency in Face Forgery)
  - Extracts frequency maps using Discrete Cosine Transform (DCT) to spot upsampling artifacts.
- **Fusion Block:** Cross-Attention Module that weighs which stream is more reliable based on image quality.

TRAINING FOR ROBUSTNESS

Strategy 1: Frequency Masking (FMSI)

- During training, we randomly "delete" parts of the frequency spectrum.
- Benefit: Forces the model to learn general forgery traces rather than memorizing specific camera noises or generator fingerprints.

Strategy 2: The Degradation Pipeline

- We do not train on clean images. We train on a "hostile" pipeline:
  - **JPEG Compression:** QF 30–60.
  - **Gaussian Blur:** Simulating out-of-focus webcams.
  - **Downscaling:** Resizing to 64px and back.

EVALUATION & METRICS

Primary Metric: AUC (Area Under Curve)

- Critical for cybersecurity to balance False Rejection Rate (locking out real users) vs. False Acceptance Rate (letting attackers in).

Anticipated Results:

- **Cross-Generator Generalization:** ~85-90% AUC on unseen models (vs. 60% baseline).
- **Robustness:** High performance maintained even at JPEG Quality 50.

CONCLUSION

Summary:

- Our Hybrid-Swin solution addresses the NTIRE 2026 challenge of generalization.
- It is designed to be future-proof against new generative models via Frequency Masking.

Real-World Application:

- **Automated e-KYC Defense:** Preventing synthetic identity fraud.
- **Disinformation Shield**: Real-time flagging of manipulated news footage.

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